

MATH 3312, SPRING 2026: SOLUTIONS TO HOMEWORK 1

(1) Determine if the following statements are true or false. Give a line or two in justification.

(a) For any commutative ring R and polynomials $f(x), g(x) \in R[x]$, we have

$$\deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x))$$

Solution. False. This is false in a ring like $\mathbb{Z}/4\mathbb{Z}$ where $2x \cdot (2x + 1) = 1$ has degree 0 even though each factor has degree 1.

(b) $\mathbb{Z}/24\mathbb{Z}$ with its usual addition and multiplication is a field.

Solution. False. 24 is not a prime.

(c) The quotient of $\mathbb{Z}[x]$ by the subgroup $\langle x \rangle$ generated by x has the structure of a commutative ring such that the quotient homomorphism

$$\mathbb{Z}[x] \rightarrow \mathbb{Z}[x]/\langle x \rangle$$

is a ring homomorphism.

Solution. False. If this existed, then, for instance, the quotient homomorphism π would satisfy

$$x^2 + \langle x \rangle = \pi(x \cdot x) = \pi(x)\pi(x) = 0 \cdot 0 = 0.$$

But, since $x^2 \notin \langle x \rangle$, this is a contradiction.

(d) There are only 3 finite abelian groups of order 8 up to isomorphism.

Solution. True. By the fundamental theorem, up to isomorphism, the three possibilities are

$$\mathbb{Z}/8\mathbb{Z}, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}.$$

(e) For a commutative ring R , the ideal generated by a polynomial $f(x) \in R[x]$ is all of $R[x]$ if and only if $f(x) = a$ is a constant polynomial with $a \in R^\times$.¹

Solution. False. For the ideal to be all of $R[x]$, we just need to know that $1 \in (f(x))$: If 1 is a multiple of $f(x)$, then every polynomial is. But in $\mathbb{Z}/4\mathbb{Z}[x]$, note that we have $(1 + 2x) \cdot (1 + 2x) = 1$: This means that $(1 + 2x)$ is all of $\mathbb{Z}/4\mathbb{Z}[x]$ but $1 + 2x$ is of course not a constant polynomial.

(2) Show that, for any commutative ring R , there is a unique ring homomorphism $\mathbb{Z} \rightarrow R$, and verify in addition that, if $R = k$ is a field, then we have the following dichotomy:

(a) Either the homomorphism $\mathbb{Z} \rightarrow k$ is injective;

(b) Or its kernel is $p\mathbb{Z}$ for a unique prime p .

Solution. The unique homomorphism is the one sending any integer m to $m \cdot 1_R$, where $1_R \in R$ is the multiplicative identity.

Suppose now that $R = k$ is a field, and write $\varphi : \mathbb{Z} \rightarrow k$ for the unique homomorphism $\varphi(m) = m \cdot 1_k$. Now, there are two possibilities for $\ker \varphi$: Either it is 0, in which case φ is injective, or it has a non-zero kernel, which has to be of the form $m\mathbb{Z}$ for some $m \neq 0$ (why?).

If m is not prime, then we can write $m = ab$ with $a, b < m$, and we would then find

$$0 = \varphi(m) = \varphi(ab) = \varphi(a)\varphi(b) \in k.$$

Since k is a field, this can only happen if one of $\varphi(a) = 0$ or $\varphi(b) = 0$ holds (why?). But $a, b \notin m\mathbb{Z} = \ker \varphi$!

This contradiction shows that the kernel is $p\mathbb{Z}$ for some (uniquely determined) prime p .

¹This is a tricky one!

Definition 1. If alternative (1) holds, we say that k has **characteristic** 0. If alternative (2) holds, we say that k has **characteristic** p .

- (3) Construct a ring R with the following property: For any other commutative ring R' , giving a ring homomorphism $f : R \rightarrow R'$ is equivalent to giving an element $x \in R'$ such that $x^2 + 1 = 0$.

Solution. The basic observation is that homomorphisms out of quotient rings are governed by the factoring triangle, and so yield *subsets* of homomorphisms of the bigger ring. Here, we know that giving a homomorphism $\mathbb{Z}[x] \rightarrow R'$ amounts to specifying an element $r \in R'$, and then just looking at the *evaluation-at- r* homomorphism

$$\text{ev}_r : \sum_i a_i x^i \mapsto \sum_i a_i r^i \in R'.$$

We now want to impose the additional condition $r^2 + 1 = 0$, which amounts to requiring that the polynomial $x^2 + 1$ is in $\ker \text{ev}_r$. The factoring triangle now tells us that this is the *same* as asking for ev_r to factor through the quotient homomorphism (why?)

$$\mathbb{Z}[x] \rightarrow \mathbb{Z}[x]/(x^2 + 1).$$

In other words, $\mathbb{Z}[x]/(x^2 + 1)$ is the R we are looking for.

Said slightly differently, we have a bijection

$$R' \xrightarrow[\simeq]{r \mapsto \text{ev}_r} \{\text{Ring homomorphisms } \mathbb{Z}[x] \rightarrow R'\}.$$

Under this bijection, the subset $\{r \in R' : 0 = r^2 + 1\}$ is carried to the set of ring homomorphisms $\text{ev}_r : \mathbb{Z}[x] \rightarrow R'$ such that $0 = r^2 + 1 = \text{ev}_r(x^2 + 1)$. But this is exactly the set of ring homomorphisms $\text{ev}_r : \mathbb{Z}[x] \rightarrow R'$ that factor through $\mathbb{Z}[x]/(x^2 + 1)$.

- (4) Fix a prime p , and construct a ring R with the following property: For any other commutative ring R' , giving a ring homomorphism $f : R \rightarrow R'$ is equivalent to giving an element $x \in R'$ such that $x^2 + x + 1 = 0$ and requiring the condition that $p \cdot 1_R = 0 \in R$.

Solution. The reasoning is the same as the previous problem, except we have to kill not just $x^2 + x + 1$, but also p as well. This gives us $R = \mathbb{Z}/p\mathbb{Z}[x]/(x^2 + x + 1) = \mathbb{F}_p[x]/(x^2 + x + 1)$.

For any commutative ring R , write $R^\times \subset R$ for the set of invertible elements in R . For $a \in R$, write $m_a : R \xrightarrow{b \mapsto a \cdot b} R$ for the function given by multiplication-by- a .

An element $a \in R$ is a *zero divisor* if there exists a *non-zero* element $b \in R$ such that $a \cdot b = 0$. Otherwise, it is a *non-zero divisor* (the hyphenation here is still controversial).

- (5) With the above notation:

- (a) Show that m_a is a homomorphism of additive groups (such a self-homomorphism is called an **endomorphism** of R as an additive group);

Solution. This is just the distributive property:

$$m_a(b + c) = a \cdot (b + c) = a \cdot b + a \cdot c = m_a(b) + m_a(c).$$

- (b) Show that m_a is invertible if and only if $a \in R^\times$ is invertible.

Solution. For m_a to be invertible, it has to in particular be surjective, meaning there exists $b \in R$ such that $ab = m_a(b) = 1$, meaning $a \in R^\times$. Conversely, if $a \in R^\times$, then $m_{a^{-1}}$ is an inverse for m_a .

- (c) Show that m_a is injective if and only if a is a non-zero divisor.

Solution. m_a is injective if and only if $a \cdot b = m_a(b) = 0$ implies that $b = 0$. This is basically the definition of being a non-zero divisor.

A ring R is an **integral domain** if it contains no zero divisors apart from 0.

(6) Prove the following:

(a) Every field is an integral domain;

Solution. We actually already used this above, and have seen this in lecture: If $a \cdot b = 0$ and $a \neq 0$, then a has an inverse a^{-1} , and we find

$$b = 1 \cdot b = (a^{-1}a) \cdot b = a^{-1} \cdot (ab) = a^{-1} \cdot 0 = 0.$$

(b) Every *finite* integral domain R is a field;

Solution. Suppose that $a \in R \setminus \{0\}$. Since it is a non-zero divisor $m_a : R \rightarrow R$ is *injective*. However, since R is *finite*, it is actually automatically *bijective*, which means that $a \in R^\times$.

(c) The following assertions are equivalent for an integer $n \geq 1$:

- (i) $\mathbb{Z}/n\mathbb{Z}$ is a field.
- (ii) $\mathbb{Z}/n\mathbb{Z}$ is an integral domain.
- (iii) n is prime.

Solution. Since $\mathbb{Z}/n\mathbb{Z}$ is finite, the equivalence of (i) and (ii) follows from parts (a) and (b). We already know that $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.

(7) Fix a finite field k , and let $f(x) \in k[x]$ be a polynomial of degree ≥ 1 . Show that the following are equivalent:

- (a) The quotient ring $k[x]/(f(x))$ is a field;
- (b) The quotient ring $k[x]/(f(x))$ is an integral domain;
- (c) $f(x)$ is irreducible.

Solution. First, note that $k[x]/(f(x))$ is in bijection with the set of polynomials of degree $< d = \deg f(x)$ with coefficients in k : This is a set with $|k|^d$ elements (why?). Again, the equivalence of (a) and (b) follows from the previous problem and the finiteness of the ring $k[x]/(f(x))$.

Suppose now that (b) holds: If $f(x)$ were *reducible*, then we would have $f(x) = a(x)b(x)$ with $\deg a(x), \deg b(x) < d$. But then the cosets $a(x) + (f(x)), b(x) + (f(x))$ would be two *non-zero* elements of $k[x]/(f(x))$ whose product is the zero coset. This is not possible by assertion (b), and so we see that $f(x)$ must be irreducible.

Finally, let's see that (c) implies one of the others. This can be done in two ways. We can do it using Bezout directly like we did for $\mathbb{Z}/p\mathbb{Z}$: If we have any $a(x)$ such that $f(x) \nmid a(x)$ (in other words, such that the coset $a(x) + (f(x))$ is non-zero), then we can find $s(x), t(x)$ such that $s(x)f(x) + t(x)a(x) = 1$, and looking at cosets now tells us that $t(x) + (f(x))$ is an inverse to $a(x) + (f(x))$ in $k[x]/(f(x))$.

We can also do this by showing (b): This also uses Bezout and boils down to the assertion

- If $f(x) \mid a(x)b(x)$, then either $f(x) \mid a(x)$ or $f(x) \mid b(x)$.

Indeed, if $f(x) \nmid a(x)$ then we have $1 = s(x)f(x) + t(x)a(x)$ as above, meaning that

$$b(x) = b(x) \cdot (s(x)f(x) + t(x)a(x)) = b(x)s(x)f(x) + t(x)a(x)b(x).$$

Now, $f(x)$ divides both summands on the right (why?) and so divides $b(x)$ as well.

(8) Let R be a commutative ring, and let $f(x) \in R[x]$ be a *monic* polynomial of degree n . Show:

(a) $a \in R$ is such that $f(a) = 0$ if and only if $f(x) = (x-a)g(x)$ for some monic polynomial $g(x) \in R[x]$;

Solution. If $f(x) = (x-a)g(x)$, then clearly $f(a)$ will be 0. For the converse, use division by the monic polynomial $x-a$ to write

$$f(x) = (x-a)q(x) + b$$

where $b \in R$ is *constant* (why is this possible?). Then $f(a) = b$ is zero precisely when $b = 0$, which is equivalent to saying that $f(x) = (x-a)q(x)$.

(b) If $R = k$ is a field, then the set $\{a \in k : f(a) = 0\}$ has at most n elements.

Solution. Use induction on n : If $n = 1$, then this is easy. Otherwise, if $f(a) = 0$, we can write $f(x) = (x - a)g(x)$ where $\deg g(x) = n - 1$. By inductive hypothesis, the set

$$\{b \in k : g(b) = 0\}$$

has at most $n - 1$ elements. Now, if we have $f(c) = 0$, then we get

$$(c - a)g(c) = 0.$$

This means that either $c = a$, or $g(c) = 0$, meaning that c is one of the at most $n - 1$ zeros of $g(x)$ in k . So $f(x)$ has at most one additional zero than $g(x)$, and so has at most n zeros.